

EXPERIMENT 3

Creation of a Simple Network Topology Using Open-Source Network Virtualization Tools

Objective

To create and configure a simple network topology using an open-source network virtualization tool and verify communication between the virtual network nodes.

Requirements / Tools

1	Ubuntu Linux Virtual Machine
2	VirtualBox
3	Mininet
4	Simple Network Topology
5	ping and basic Linux networking commands

Procedure

1. Start the Ubuntu Linux Virtual Machine in VirtualBox.
2. Open the terminal and start Mininet.
3. Create the simple network topology with the default Mininet topology.
4. Check the available virtual nodes using the **nodes** command.
5. Check the network connections using the **net** command.
6. Verify communication between the virtual hosts using **pingall** and host-to-host ping.
7. Stop the network after completing the experiment.

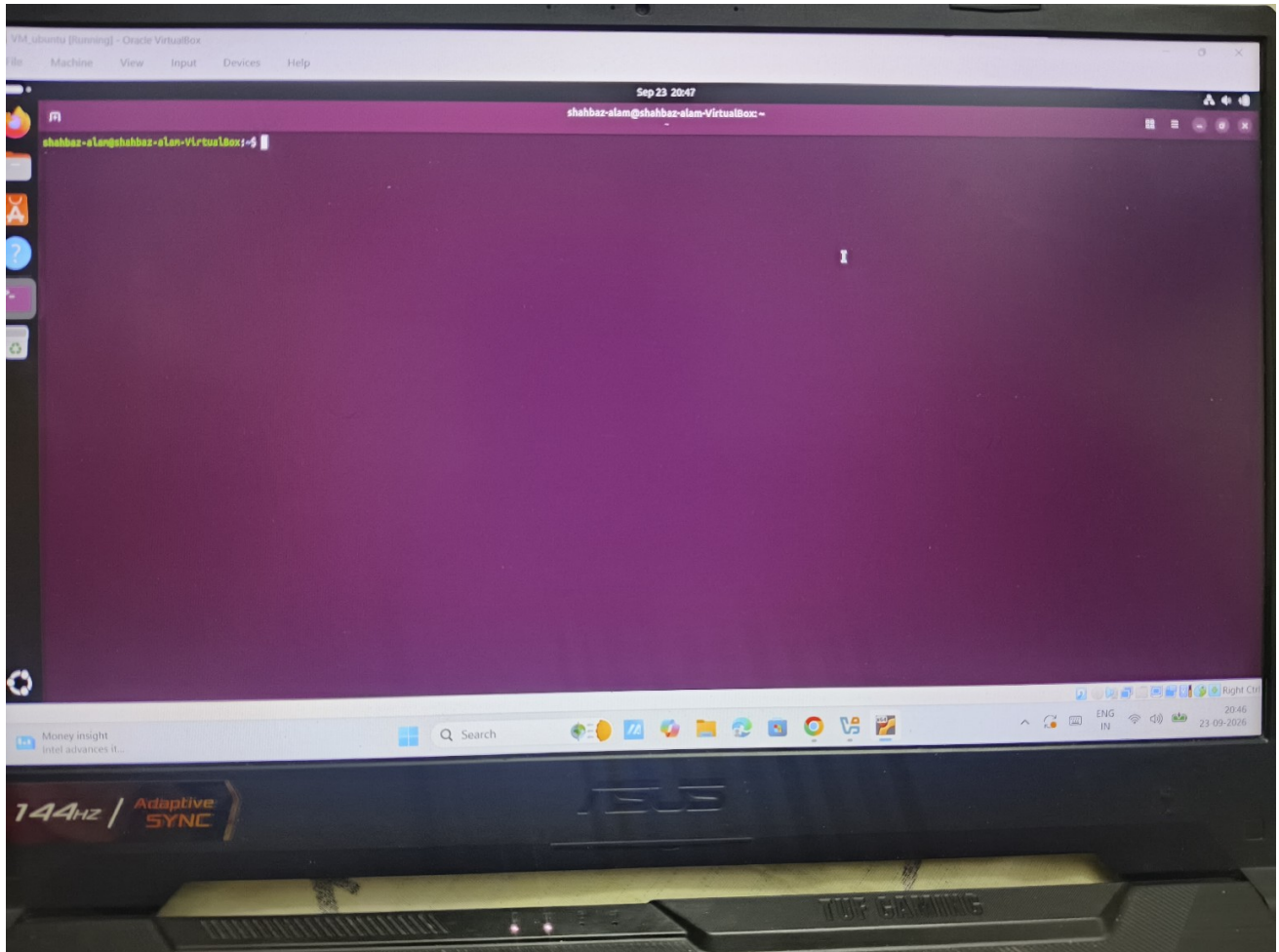
Commands Used

```
sudo mn
nodes
net
pingall
h1 ping h2
exit
sudo mn -c
```

Observation / Output

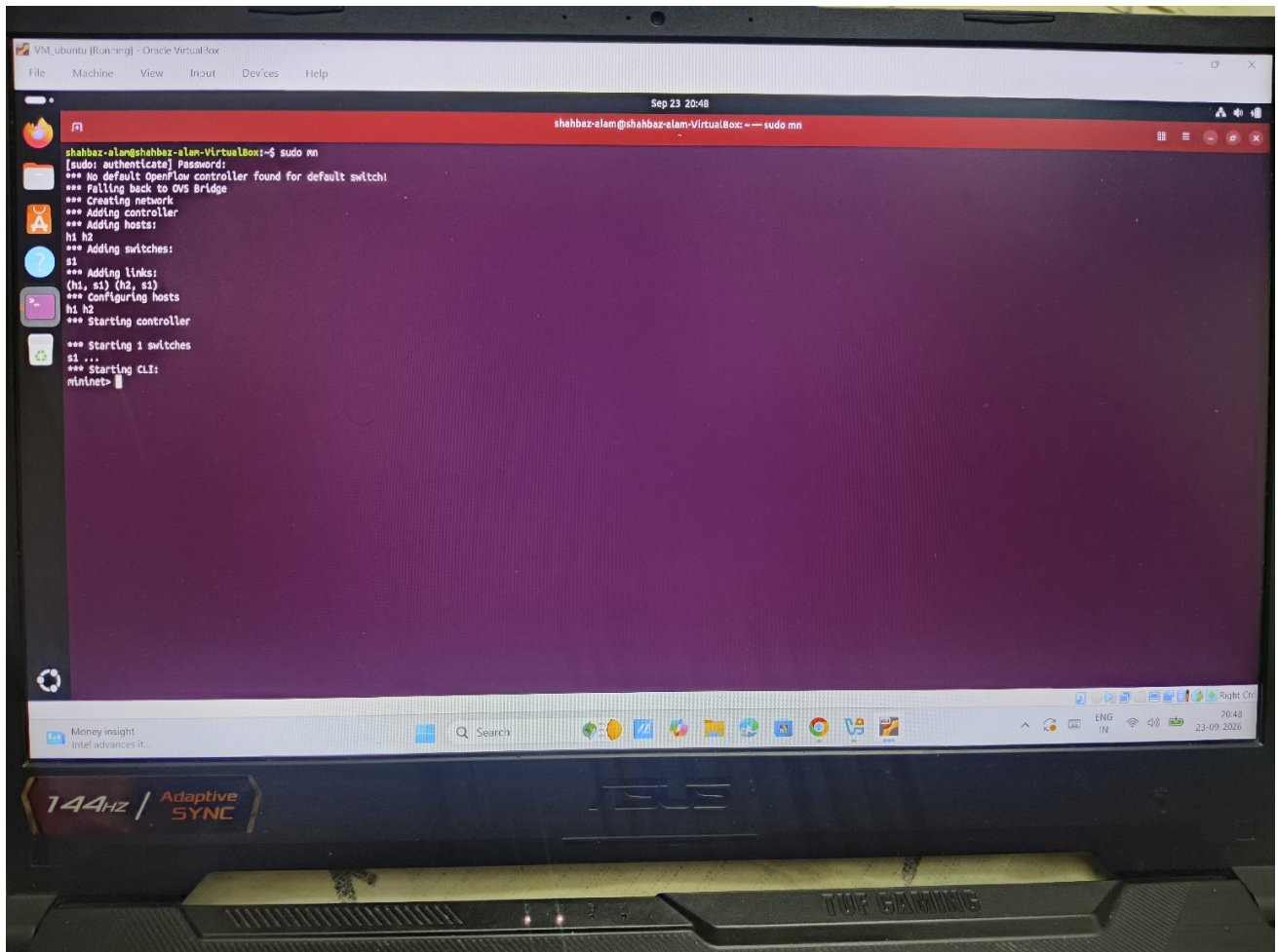
The Mininet topology contains two hosts (h1 and h2) connected through one switch (s1). The screenshots below show topology creation, node and link information, connectivity verification, and network shutdown.

Figure 1



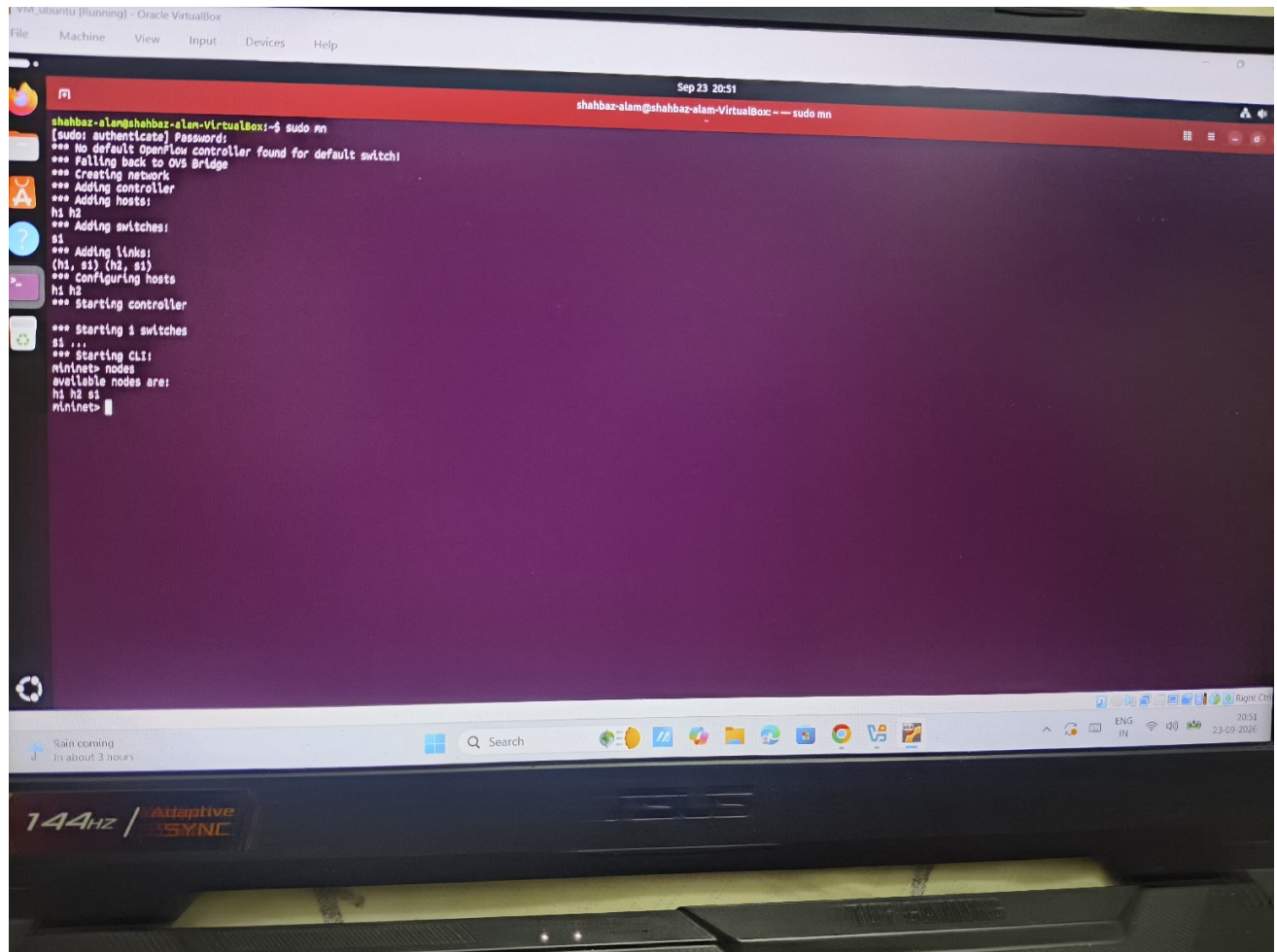
Mininet topology creation using sudo mn.

Figure 2



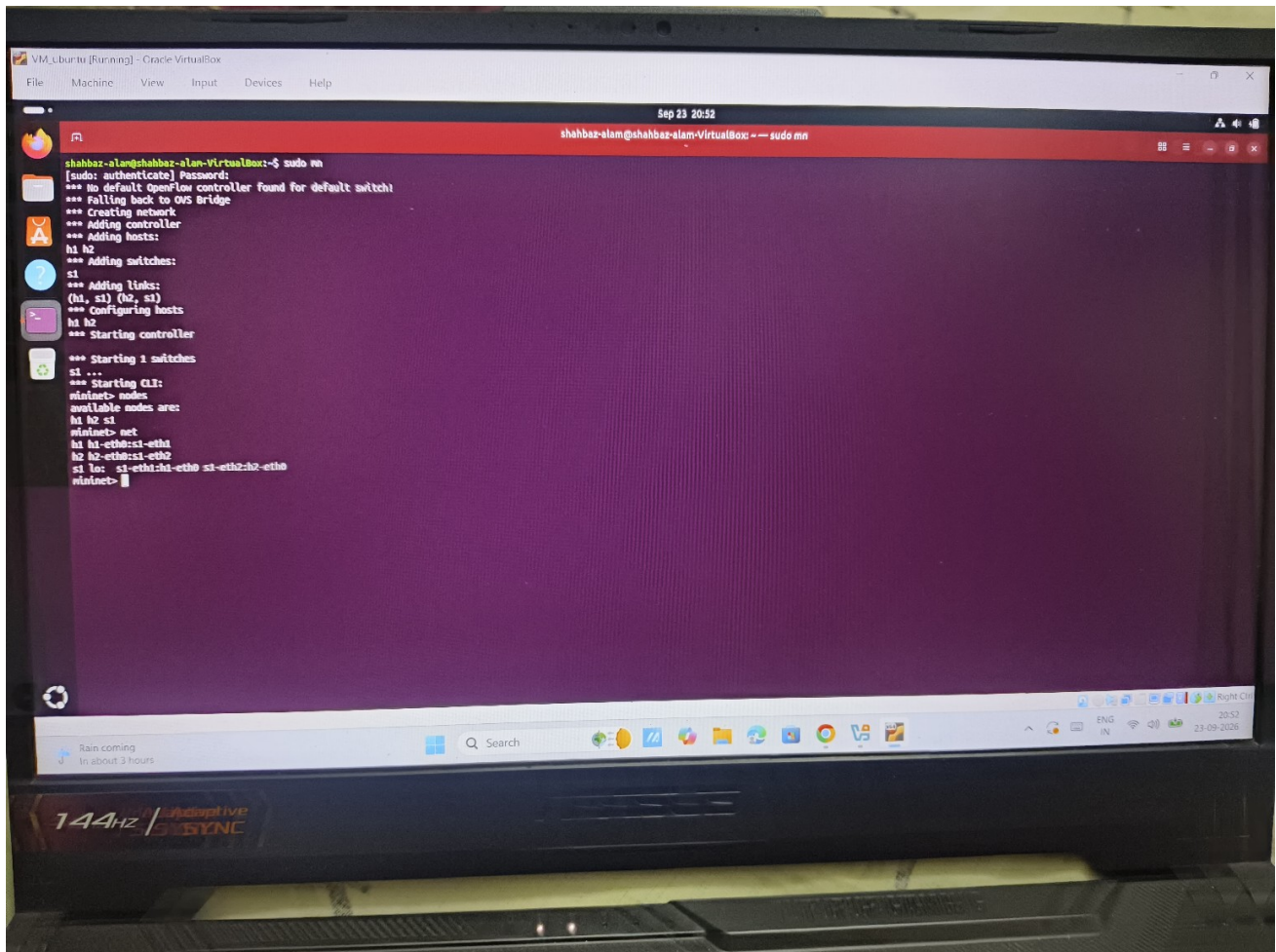
The Mininet CLI is started and ready for commands.

Figure 3



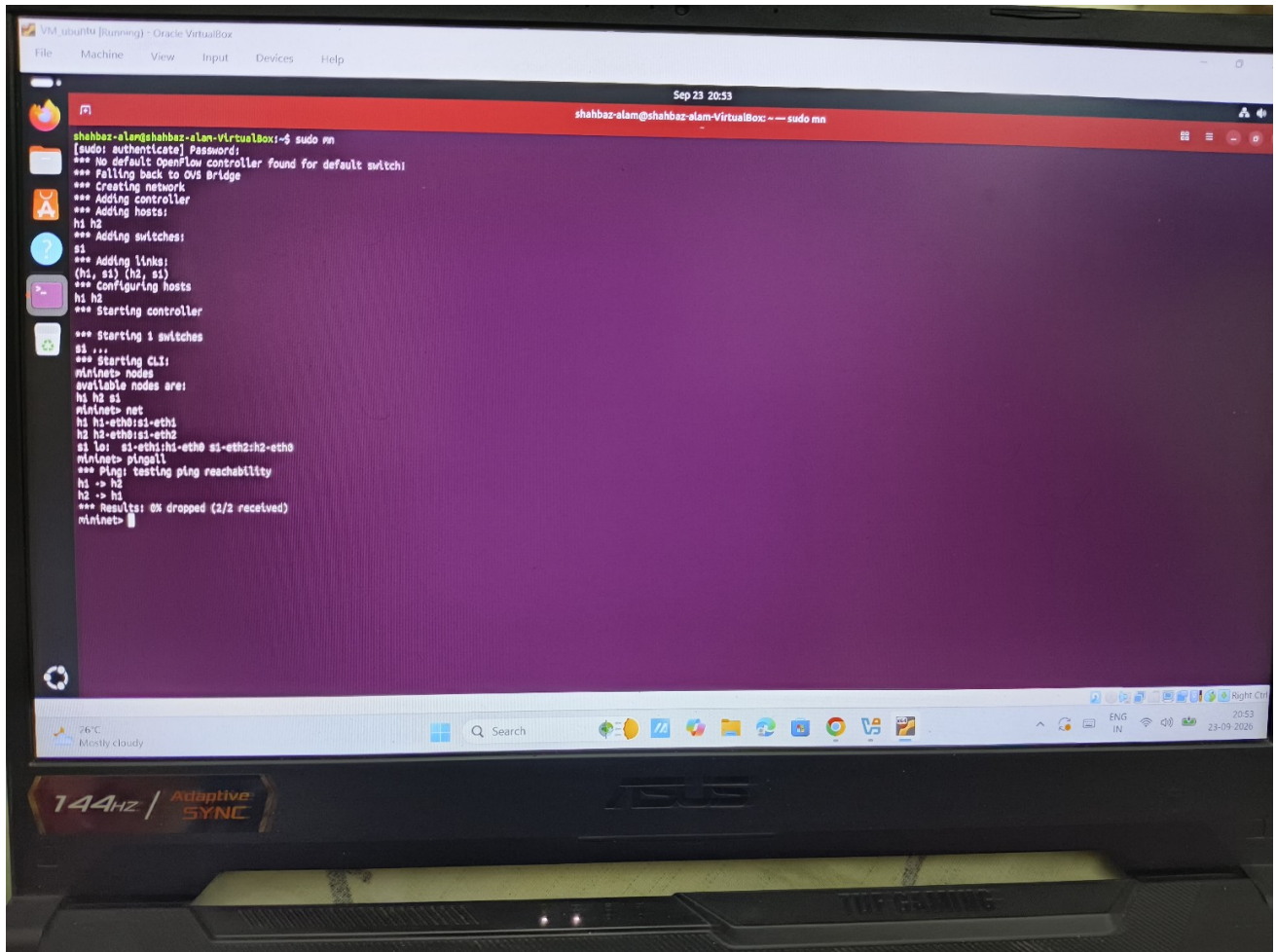
nodes command showing h1, h2 and s1.

Figure 4



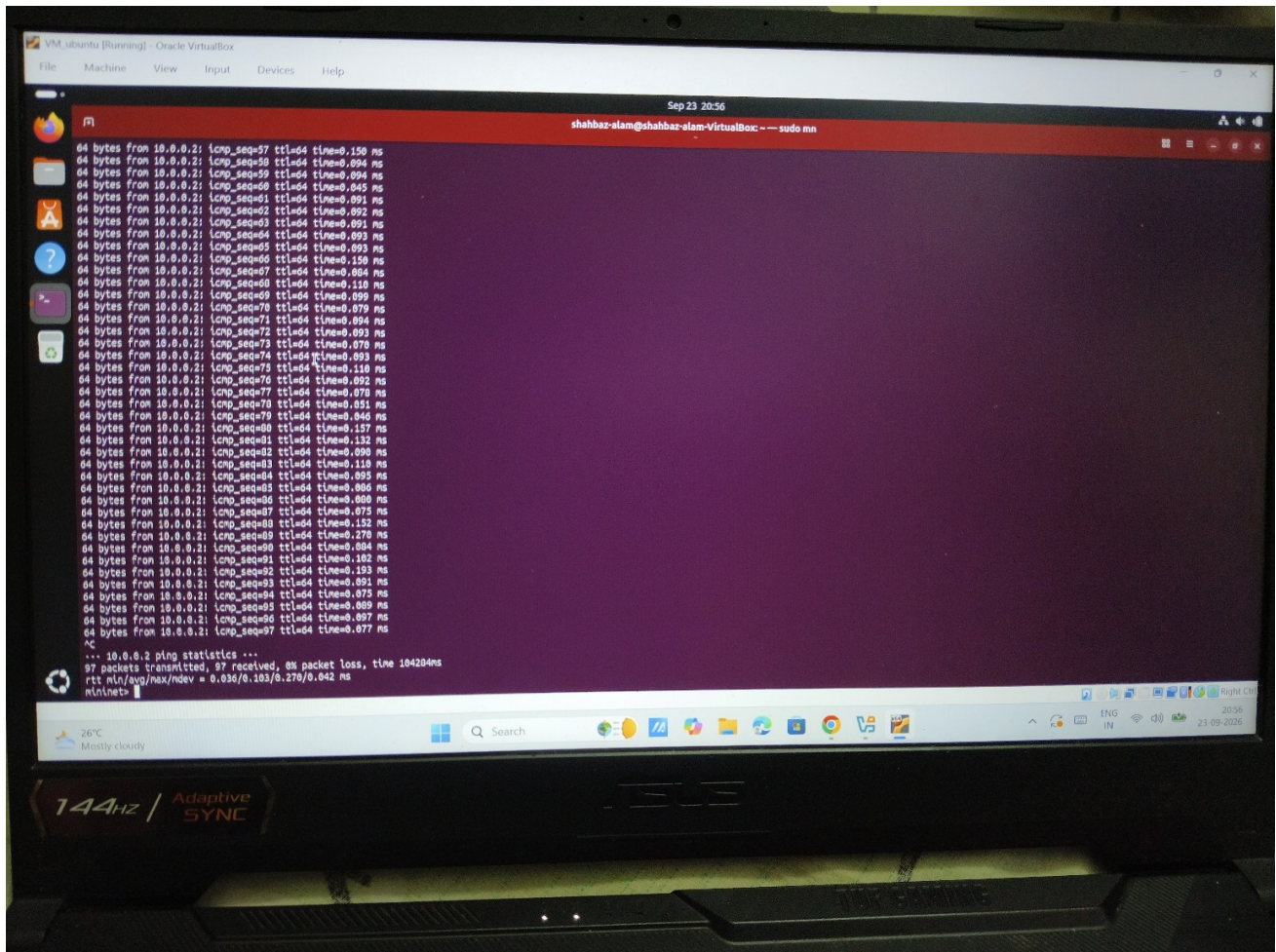
net command showing the links between h1, h2 and s1.

Figure 5



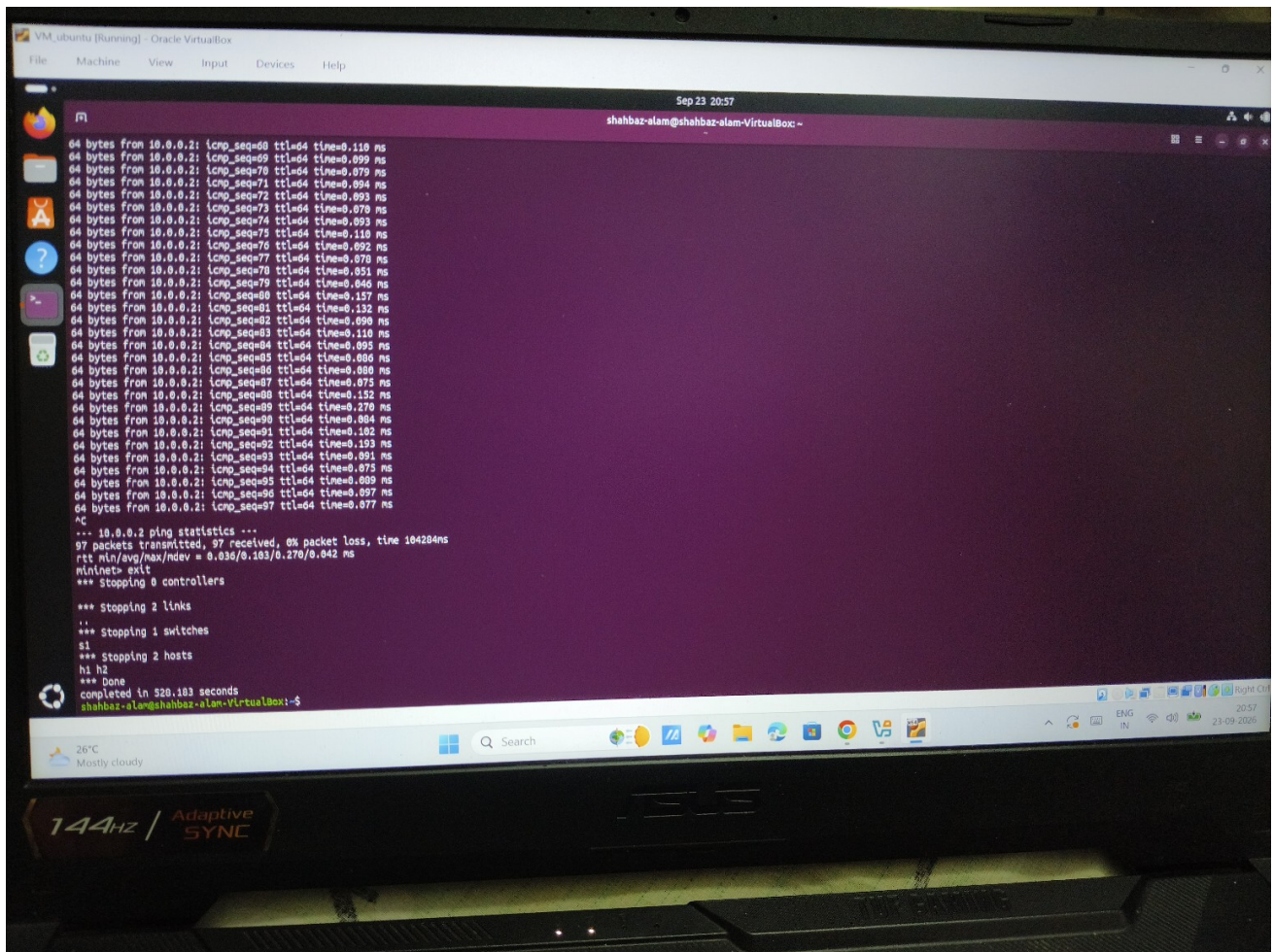
pingall output showing 0% packet loss and successful reachability.

Figure 6



Ping statistics showing successful packets and 0% packet loss.

Figure 7



Mininet network stopped and terminal returned to the shell prompt.

Result

A simple network topology was successfully created and configured using Mininet. The virtual nodes were identified, their network connections were displayed, and communication between the virtual hosts was verified using ping. The network was stopped after completion.

Conclusion

The experiment demonstrates the creation of a basic virtual network topology and verification of communication between its virtual nodes using open-source network virtualization tools.